

Commvault Professional

Course Guide

Module 3 – Security

Commvault Platform Release 2023E

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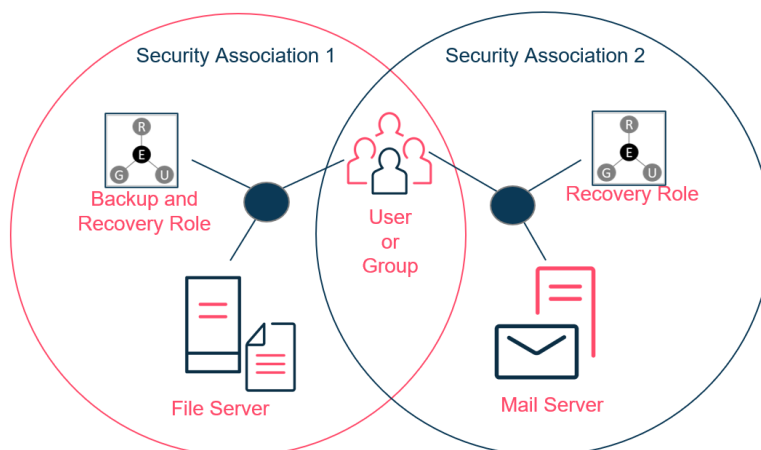
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User Security

Role-based Security

Note: Users and roles must be defined before security associations can be established.



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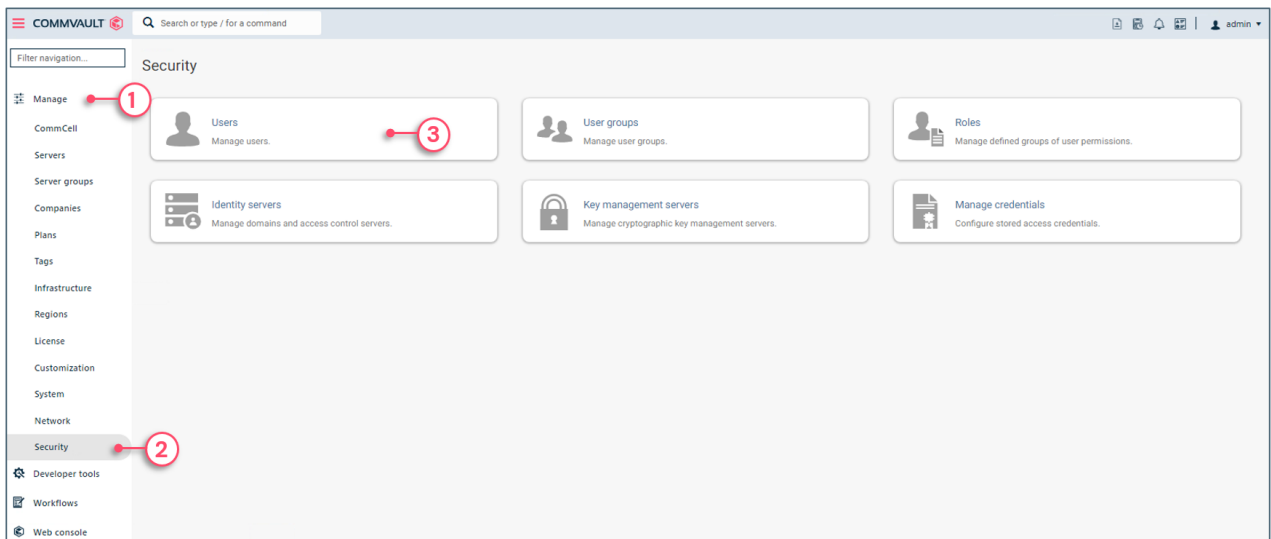
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Role-based security defines the resources and tasks that are made available to users. Security associations must be created between users or user groups, a role, and an entity.

- **User** – A CommCell user, an external domain user, a CommCell user group, or an external domain user group
- **Role** – A collection of permissions that defines the level of access granted to a user or a user group
- **Entity** – A CommCell resource such as a server, server group, plan, etc.

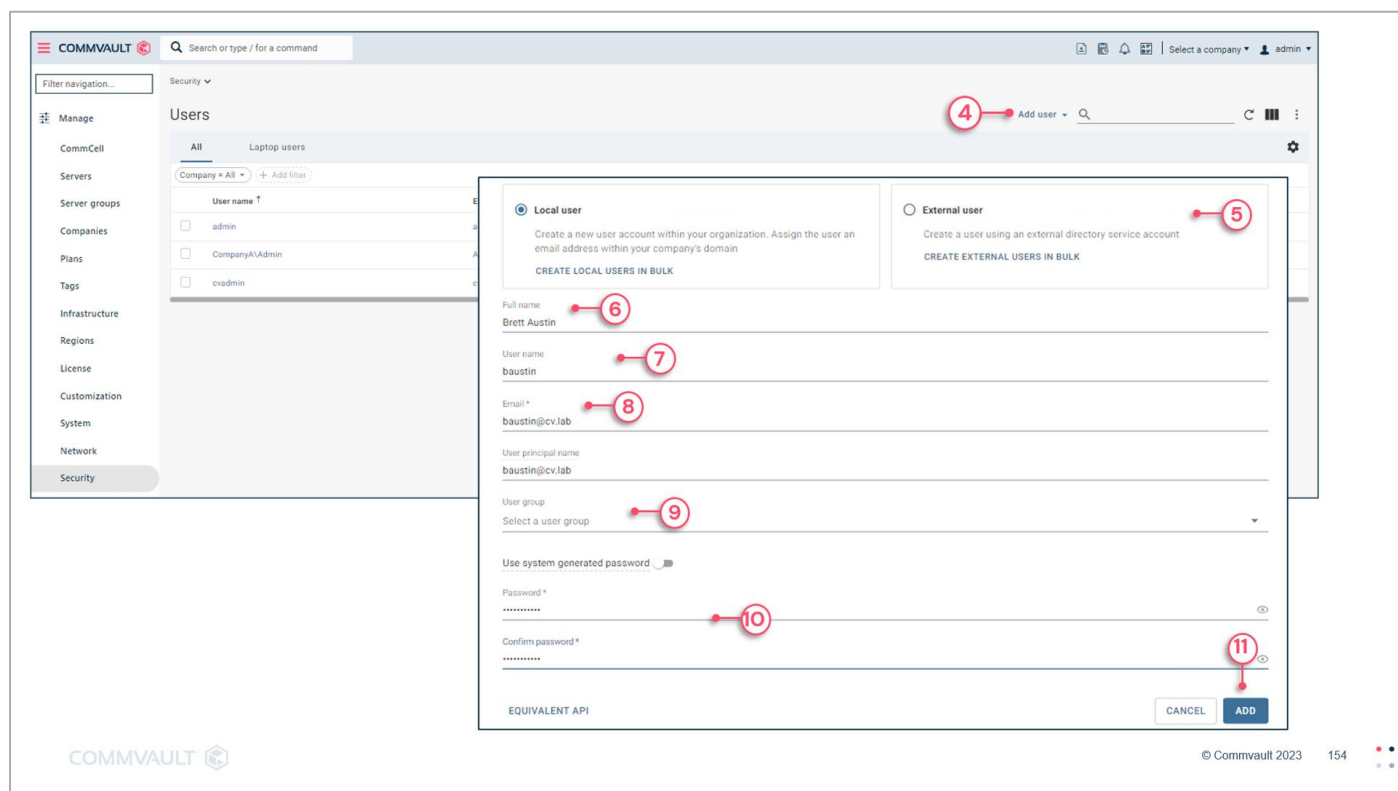
Example: A user requires backup and recovery permissions for a file server. The same user requires restore only permissions for a mail server. The user is associated with the file server entity and assigned the backup and recovery role. The same user is assigned to the mail server entity with the recovery role.

Create a User



1. In the *Navigation menu*, select **Manage**
2. Select **Security**
3. Click **Users** to manage CommCell users

Create a user ...cont'd



4. Click **Add user**
5. Choose **Local User** (CommCell) or **External User** (e.g., Active Directory) user
6. Enter the user's full name
7. Enter a login name for the user
8. Enter the user's email address
9. Select the user's group
10. Enter and confirm the password for the user
11. Click **ADD** to create the user

Tip!

When working with contractors, create individual accounts for each person so they can work in the environment, and once they leave, disable the accounts. Individual accounts are a good practice as Commvault tracks activity for each user.

Disable a User Account

The image consists of two screenshots from the Commvault web interface, illustrating the steps to disable a user account. Red circles with numbers 1, 2, and 3 highlight the specific actions.

Screenshot 1: Users List

The top screenshot shows the 'Users' page. The left sidebar has 'Security' highlighted (1). The main content area shows a table of users. The user 'baustin' is selected (2).

User name	Email	Full name	User principal name	Actions
admin	administrator@cv.lab	Administrator	Not Applicable	
baustin	baustin@cv.lab	Brett Austin	baustin@cv.lab	
CompanyAAdmin	Admin@TenantA.com	TenantA Admin	Admin@TenantA.com	
cvadmin	cvadmin@cv.lab	cvadmin	cvadmin@cv.lab	

Screenshot 2: User Summary

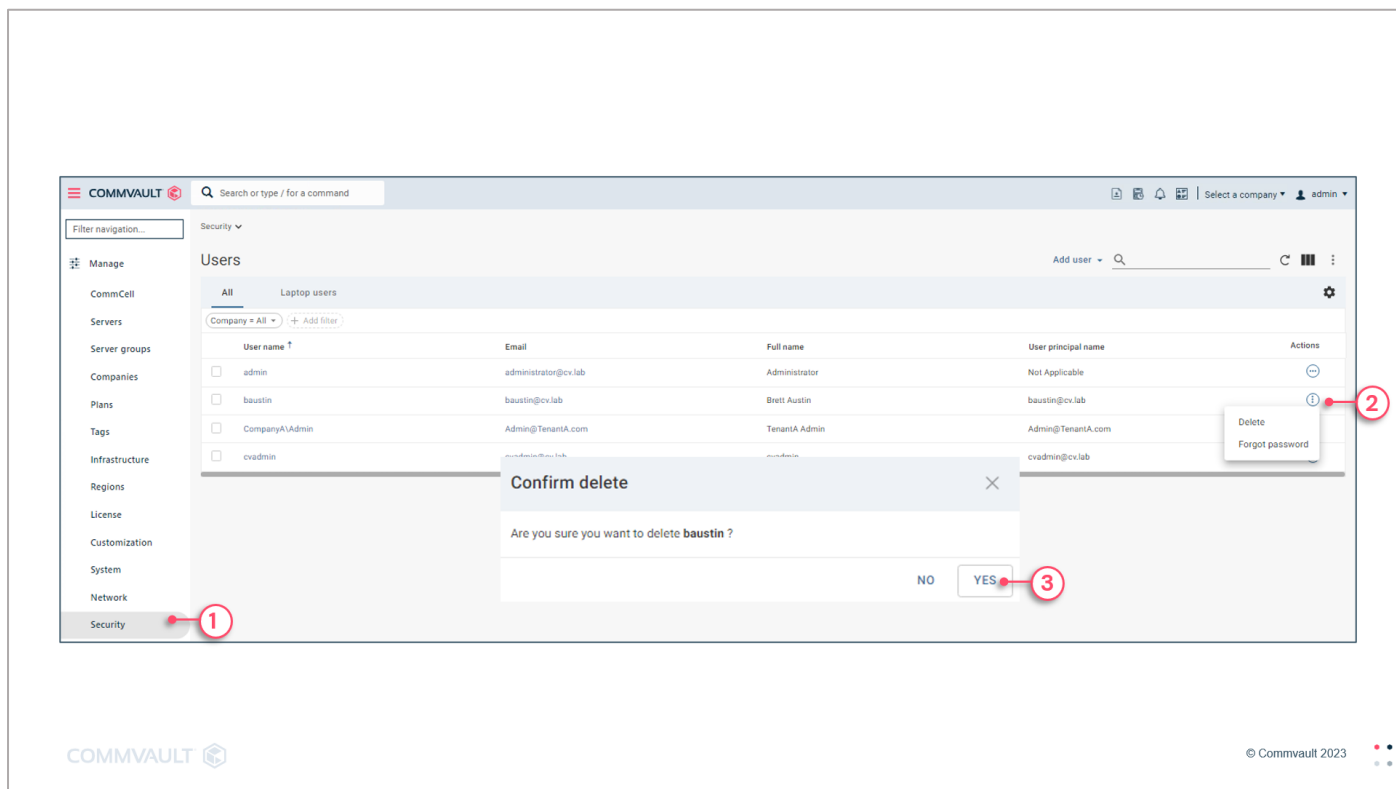
The bottom screenshot shows the 'baustin' user summary page. The 'Disable' button is highlighted (3).

User summary

Full name	Email	Authentication method	User principal name
Brett Austin	baustin@cv.lab	Commvault	baustin@cv.lab

1. Select **Manage > Security > Users**
2. Click the username
3. Click **Disable** to disable the account

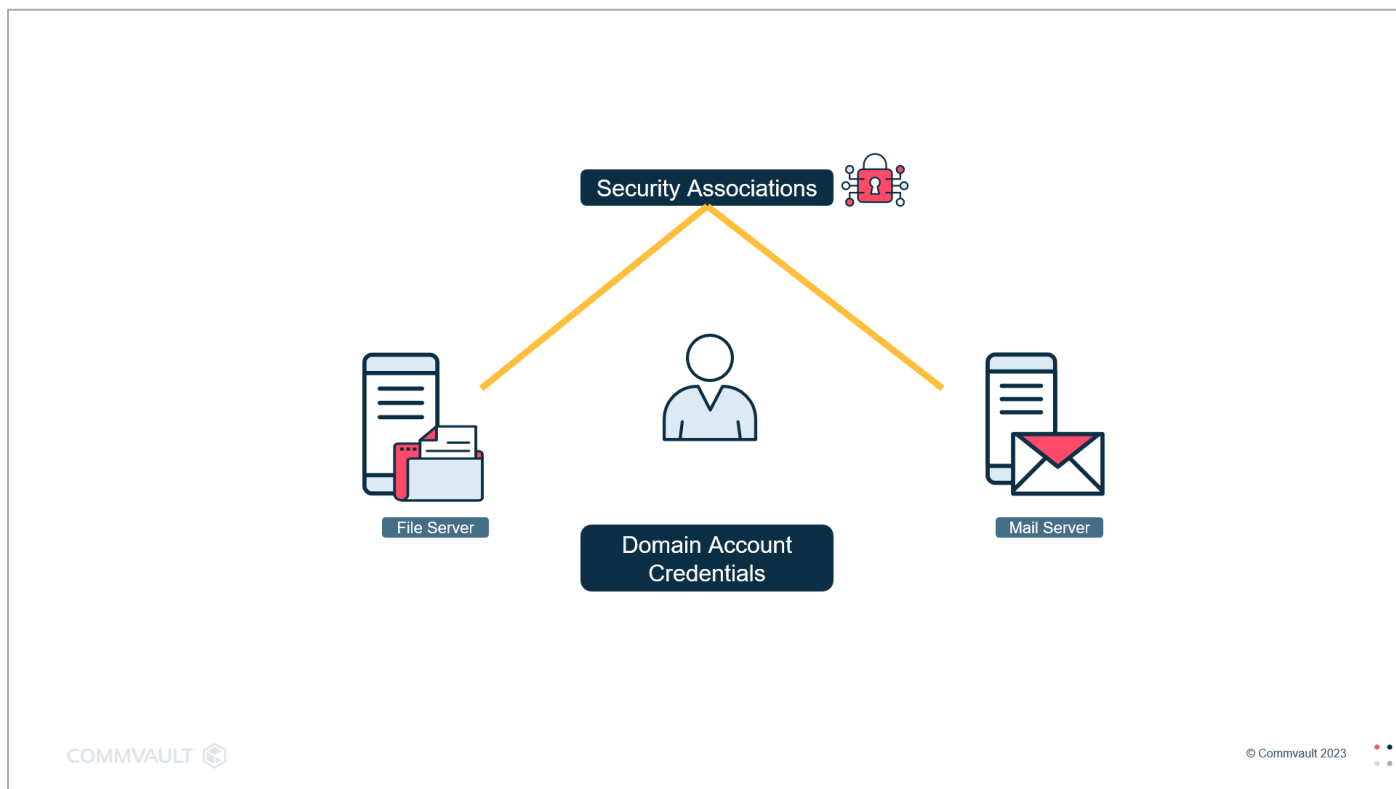
Delete a User Account



Deleting an account prevents the user from any future connection in the Commvault environment. Any entities or configurations created by the user remain intact.

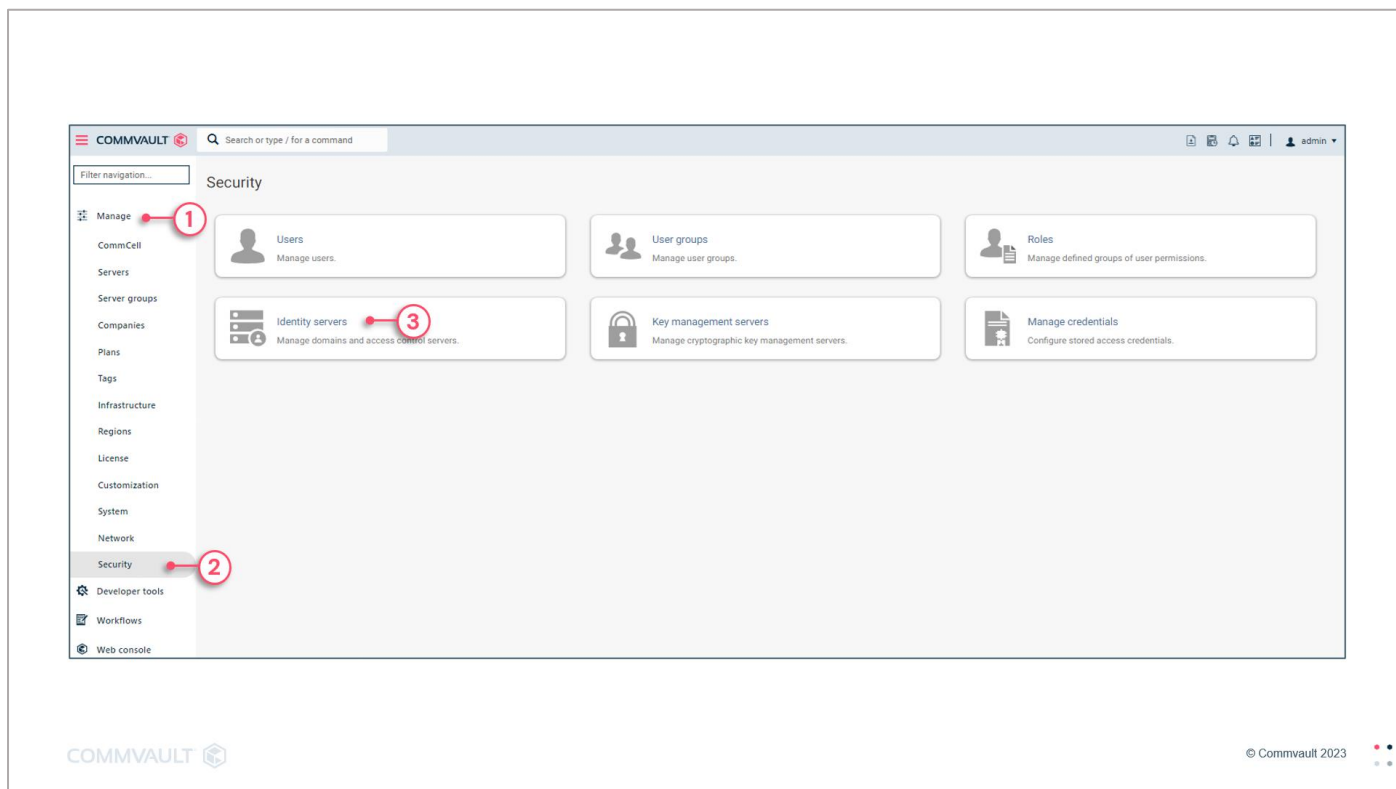
1. Select **Manage > Security > Users**
2. Click the **Actions icon** and select **Delete**
3. Click **Yes** to confirm delete

External Domains



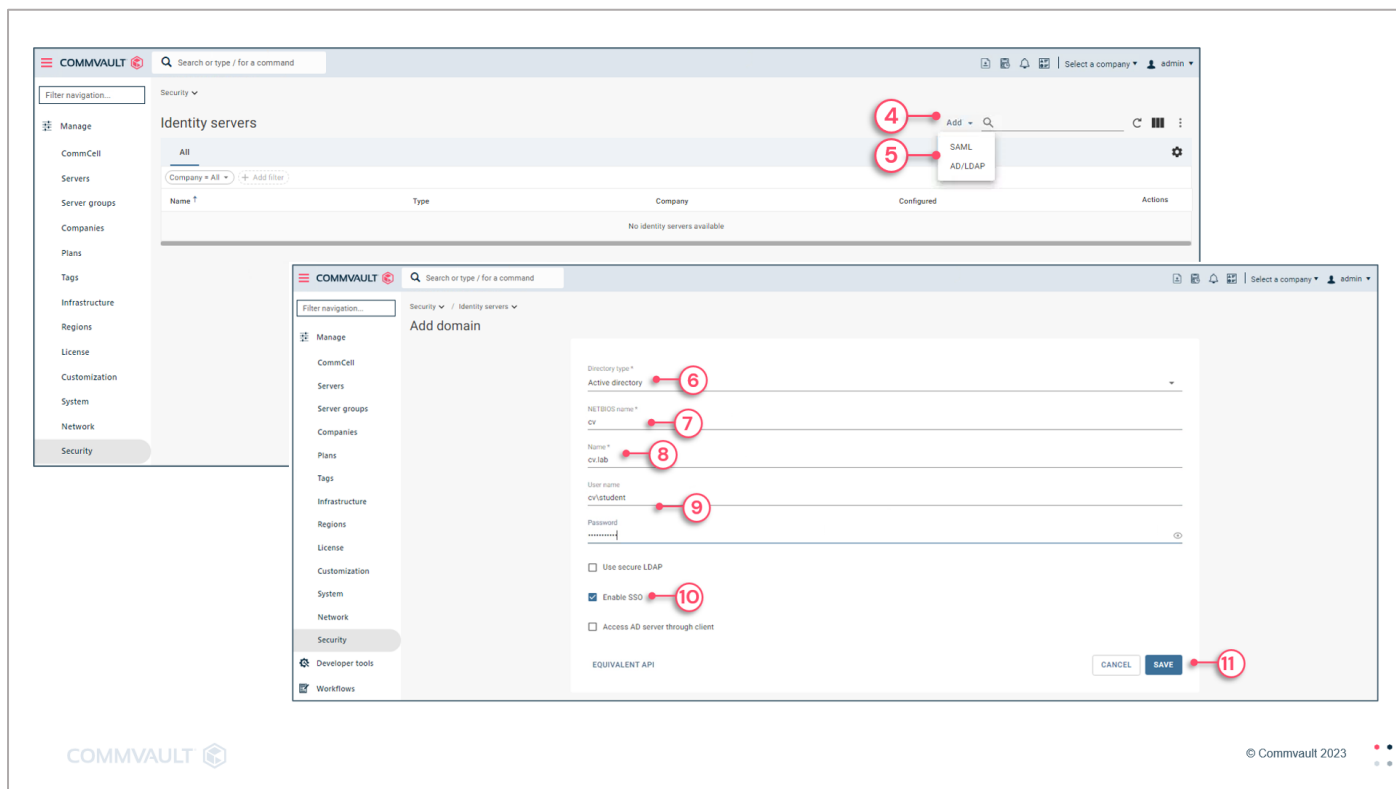
External domain users are used to define security associations in Commvault software. This allows a user to use their regular domain account, or connect via single sign-on, so that they do not need to remember additional login and password details. Commvault supports SAML and LDAP.

Add an External Domain



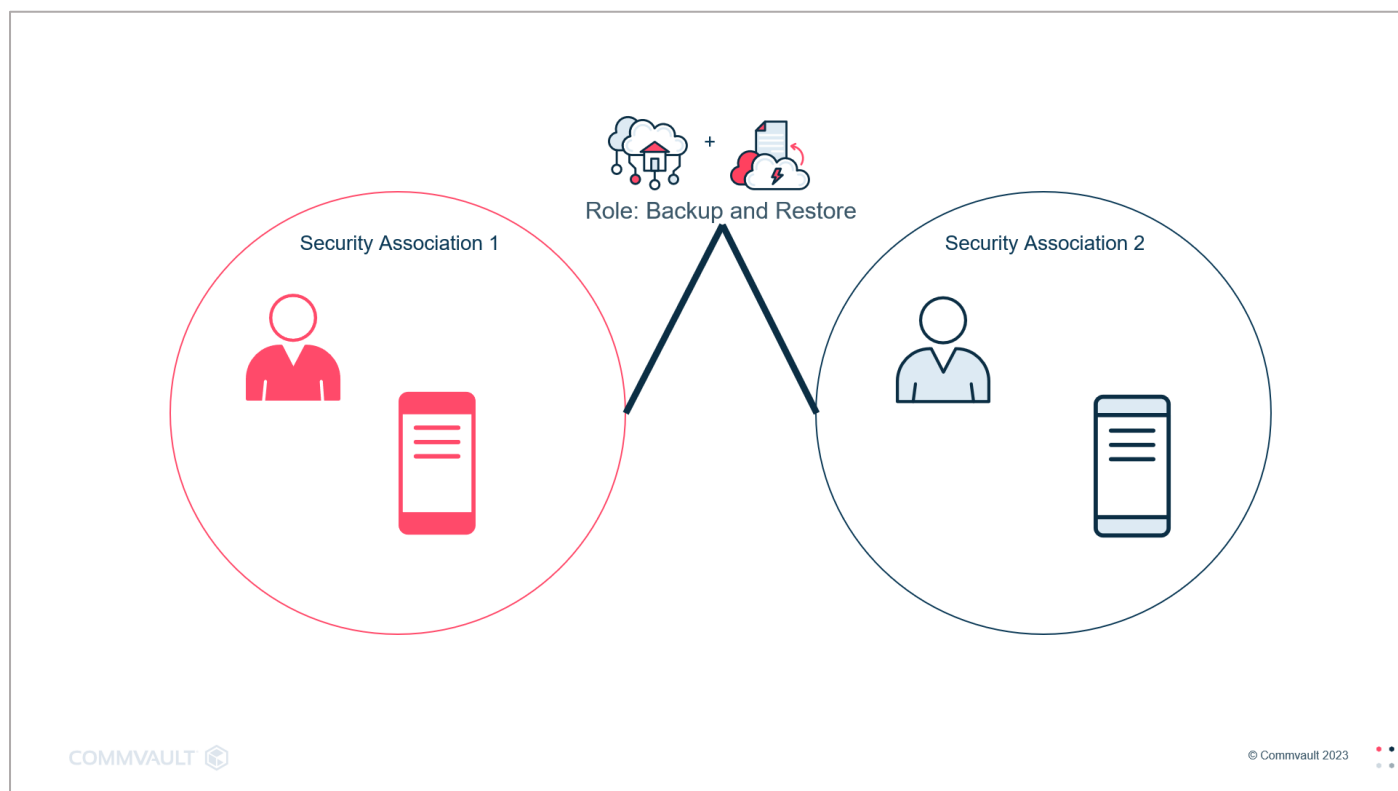
1. Select **Manage**
2. Select **Security**
3. Click the **Identity servers** tile

Add an External Domain ...cont'd



4. Click **Add**
5. Select **LDAP** or **SAML** authentication; this example uses LDAP
6. Select the LDAP directory type
7. Enter the NETBIOS domain name
8. Enter the full domain name
9. Enter the username and password used to query the domain controller
10. Check the box to **Enable SSO**; leave unchecked to keep single sign-on disabled
11. Click **SAVE** to create the connection to the domain

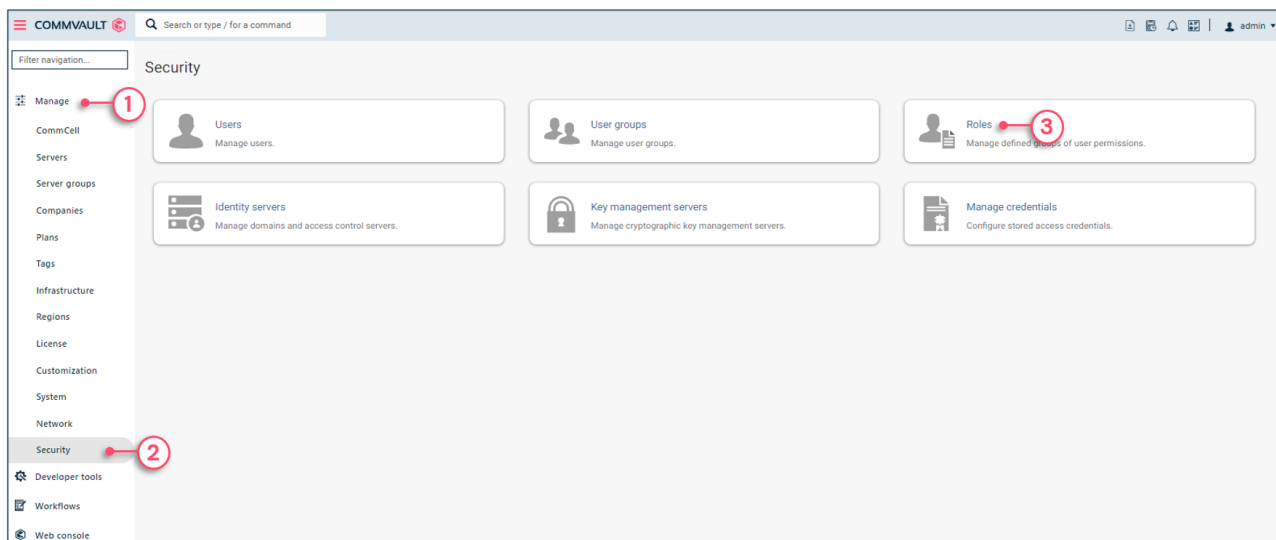
Roles



A role is a consolidated set of permissions used when creating a security association. A role can be part of as many security associations as needed, but each security association can only have one role.

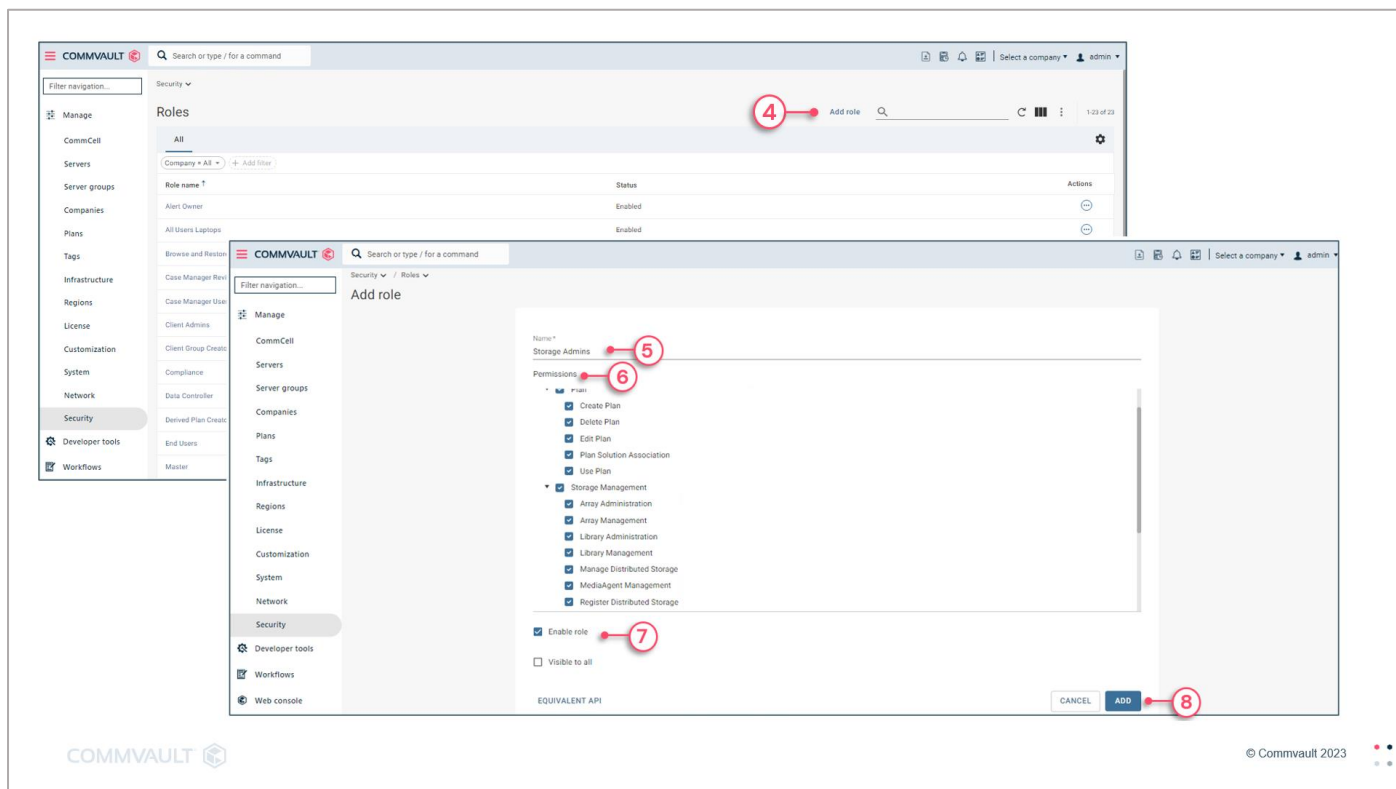
Example: If User01 requires backup and restore permissions on server A, and User02 requires backup and restore permissions on server B, you can create a single role called 'Backup and Restore' with the appropriate permissions set. This role can then be used on two different security associations, one for User01 and one for User02.

Create a Role



1. Select **Manage**
2. Select **Security**
3. Click the **Roles** tile to manage roles

Create a role ...cont'd



4. Click **Add role**
5. Enter a name; existing roles are displayed in the main window
6. Select the required permissions
7. Check the box for **Enable role**
8. Click **ADD** to create the role

Edit a Role

The screenshot displays the Commvault Roles management interface. The top panel shows a list of roles with columns for Role name, Status, and Actions. The 'Alert Owner' role is selected, and the 'Edit' action is highlighted with a red circle and the number 1. The bottom panel shows the 'Add role' configuration window for the 'Storage Admins' role. The 'Permissions' section is expanded, and the 'Show selected' checkbox is unchecked with a red circle and the number 2. The 'Alert' section is expanded, and the 'Create Alert' checkbox is checked with a red circle and the number 3. The 'Enable role' checkbox is checked with a red circle and the number 4. The 'ADD' button is visible at the bottom right of the configuration window.

1. For the desired role, click the **Actions icon > Edit**

2. Uncheck **Show selected** to display all permissions

3. Modify the required permissions

4. Click **ADD** to save modifications

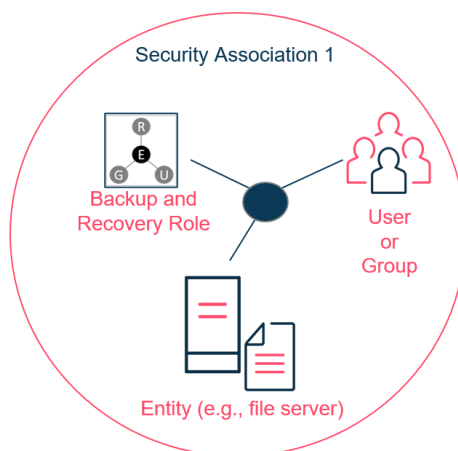
Delete a Role

The screenshot displays the Commvault Roles management page. On the left, a sidebar lists various management categories, with 'Security' selected. The main area shows a table of roles. The 'Alert Owner' role is highlighted. A context menu is open over the 'Alert Owner' row, showing 'Edit' and 'Delete' options. A red circle with the number '1' points to the 'Delete' option. A 'Confirm delete' dialog box is open in the foreground, asking 'Are you sure you want to delete Alert Owner?'. The dialog has 'NO' and 'YES' buttons. A red circle with the number '2' points to the 'YES' button.

Role name ↑	Status	Actions
Alert Owner	Enabled	[Info] [Edit] [Delete]
All Users Laptops	Enabled	[Info]
Browse and Restore	Enabled	[Info]
Case Manager Reviewer	Enabled	[Info]
Case Manager User	Enabled	[Info]
Client Admins	Enabled	[Info]
Client Group Creator	Enabled	[Info]
Compliance	Enabled	[Info]
Data Controller	Enabled	[Info]
Derived Plan Creator Role	Enabled	[Info]
End Users	Enabled	[Info]
Master	Enabled	[Info]
MSP Subscription	Enabled	[Info]

1. For the desired role, click the **Actions icon** and select **Delete**
2. Click **YES** to confirm the deletion of the role

Configure Security Associations



To use role-based security you must create a security association between users or user groups, a role, and CommCell entities. The entity defines the object or group of objects on which the defined user or users can execute tasks defined by the role. For instance, if a user needs to achieve tasks on a server, create the security association on the server entity. If the user needs to execute tasks on several servers, a server group can be leveraged on which the security association can be defined.

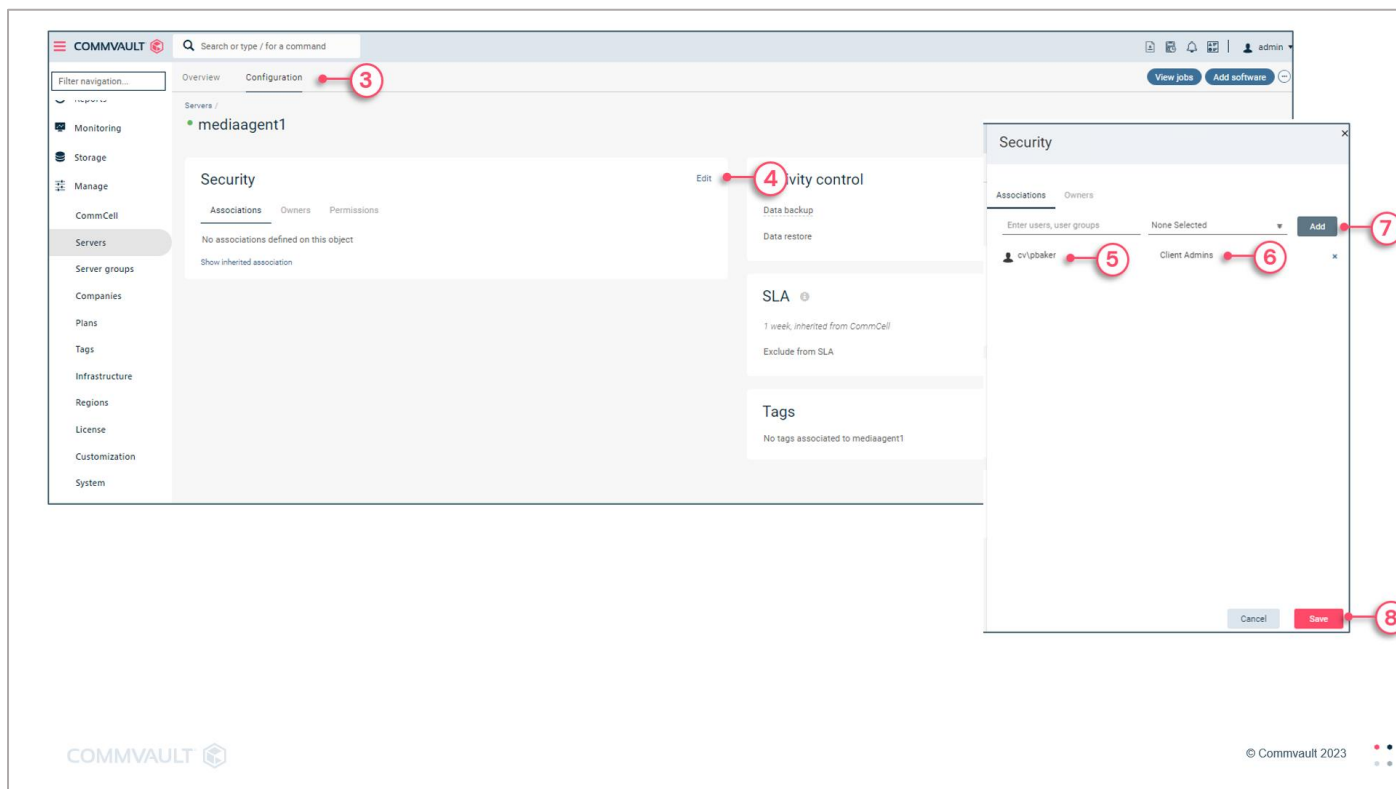
Configure Security Associations ...cont'd

The screenshot shows the Commvault web interface. On the left sidebar, under the 'Manage' section, the 'Servers' option is highlighted with a red circle and the number '1'. The main content area is titled 'Servers' and contains a table of server entities. The table has columns for Name, Agent, Roles, Version, OS, Update status, Tags, and Actions. Three rows are visible: 'csprod' (Windows File System agent), 'hyperv1' (Virtual Server agent), and 'mediaagent1' (Storage MediaAgent agent). The 'mediaagent1' row is highlighted with a red circle and the number '2'.

Name ↑	Agent	Roles	Version	OS	Update status	Tags	Actions
csprod	Windows File System	Web Server, Command Center	11.32.8	Windows Server 2022 Standard	Up to date	No tags	⋮
hyperv1	Virtual Server	MediaAgent	11.32.8	Windows Server 2022 Standard	Up to date	No tags	⋮
mediaagent1	Storage MediaAgent		11.32.8	Windows Server 2022 Standard	Up to date	No tags	⋮

1. Select **Manage > Servers**
2. Click the **CommCell entity** for which you want to create a security association

Configure Security Associations ...cont'd



3. Click the **Configuration** tab
4. In the *Security* section, click **Edit** to add a security association
5. Search for the individual username, user group, domain user or domain user group
6. Select the role
7. Click **Add** to add the security association
8. Click **SAVE** to create the security association

View Security Associations

The image displays two screenshots of the Commvault web interface, illustrating the steps to view security associations for a specific entity.

Top Screenshot: The 'Servers' menu item is selected in the left navigation pane (indicated by a red circle with the number 1). The 'Security' section is active, and the 'mediaagent1' entity is selected (indicated by a red circle with the number 2). The 'Show inherited association' link is highlighted (indicated by a red circle with the number 3).

Bottom Screenshot: The 'Show inherited association' link has been clicked, and the 'Security' section now displays a table of associations for 'mediaagent1'. The table is titled 'Security' and has columns for 'User/Group' and 'Role (Defined in)'. The table lists the following associations:

User/Group	Role (Defined in)
cvlpbaker	Client Admins
master	Master (coprod)
master	Master (infrastructure)

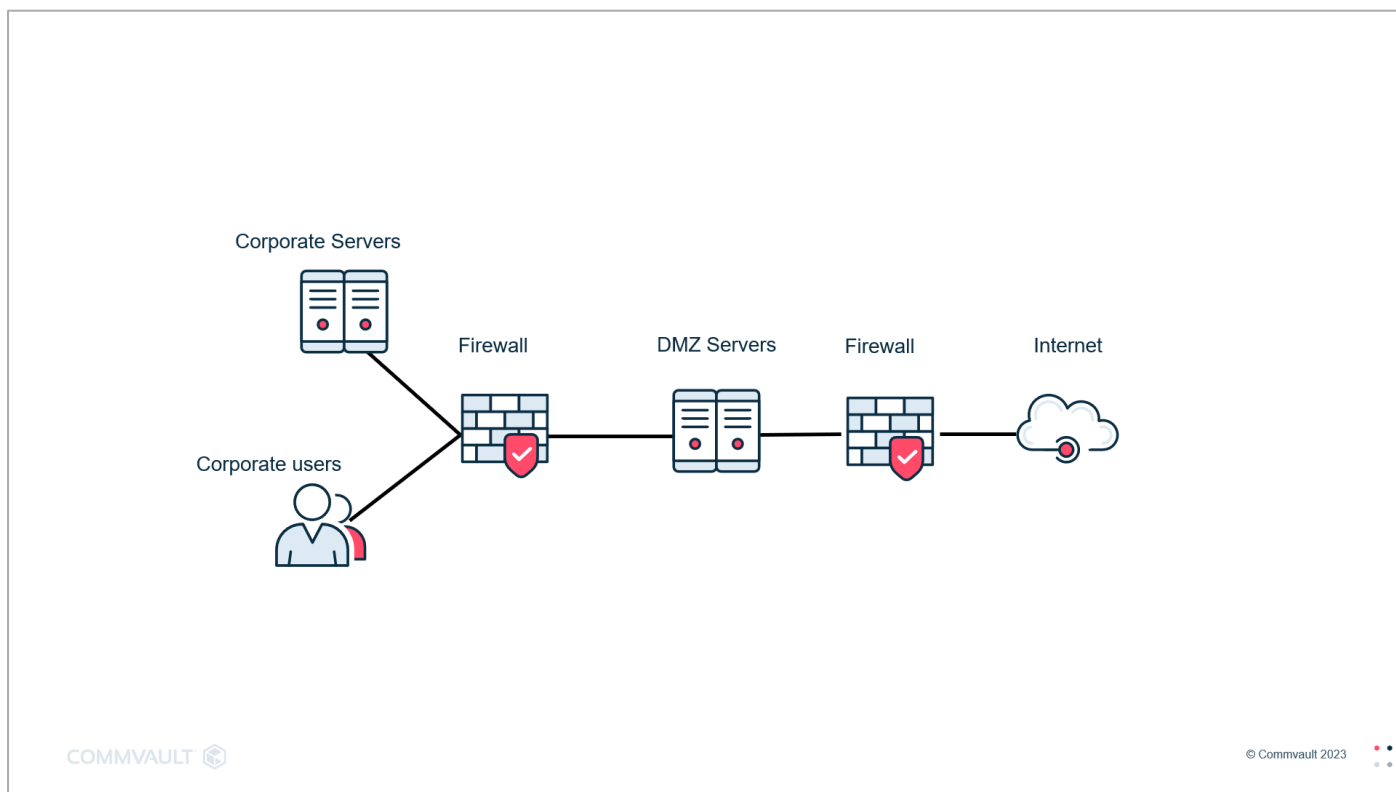
The table is highlighted with a red border. The 'Activity control' section on the right shows 'Data backup' and 'Data restore' toggles. The 'SLA' section shows '1 week, inherited from CommCell' and 'Exclude from SLA' toggle. The 'Tags' section shows 'No tags associated to mediaagent1'.

1. Select **Manage > Servers**
2. In the *Security* section, click the **name of the entity**
3. Click **Show inherited associations**

Security associations are displayed.

Network Topologies

Default Ports



Firewalls are commonly used to block TCP and UDP ports between two networks to restrict unauthorized users from accessing sensitive resources and data. An example is computers that are required to have direct access to the internet, such as web servers. These servers are usually isolated in a dedicated network called a demilitarized zone, or DMZ. This ensures that if a hacker successfully breaks into and accesses a web server, they cannot reach the corporate network since the machine is isolated in the DMZ. There are two methods of implementing network routes.

- Physical network appliance that segregates networks through physical connections
- Software-based firewall that restricts inbound and/or outbound traffic

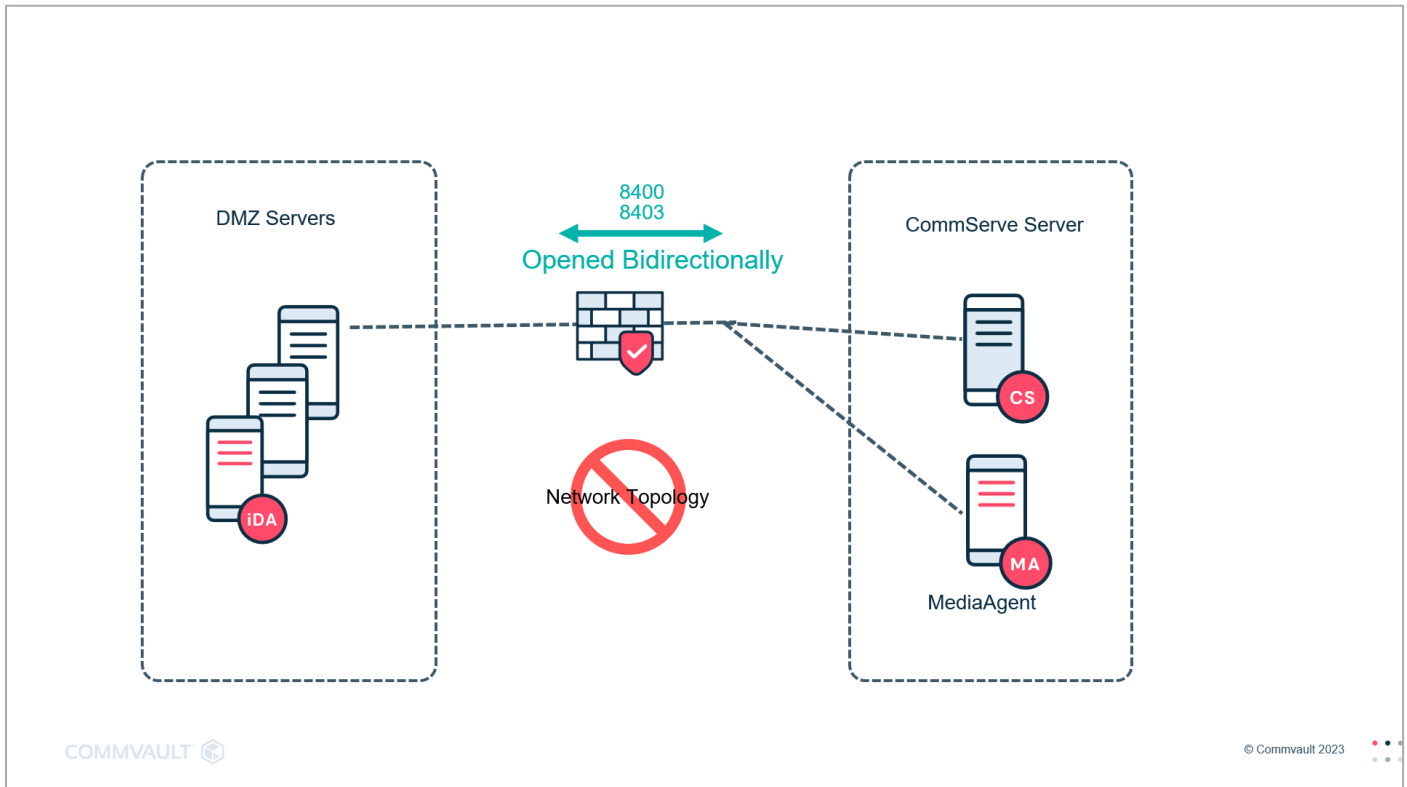
Commvault software uses predefined static and dynamic ports to handle communication and data transfer.

Using multiple ports, especially dynamic ports, presents challenges for protecting a computer behind a firewall if no other mechanism is in place. If this is the case, you would have to ask your network team to open ports 1024 to 65535 between all clients in the DMZ and the internal servers.

Default Ports:

Service	Port Number	Protocol
Commvault Communications Service	8400	TCP
Commvault Server Event Manager, GxEvMgrS service, available on CommServe server	8401	TCP
GxCVD service dynamically uses free ports for communication during data protection and data recovery jobs	1024 to 65535	TCP

Automatic Tunneling



Commvault components communicate using the traditional communication port and dynamic ports. If the system notices that the dynamic ports are blocked and therefore unavailable, it automatically encapsulates data transfers through a tunnel port. There is no need to configure any network topologies or network routes.

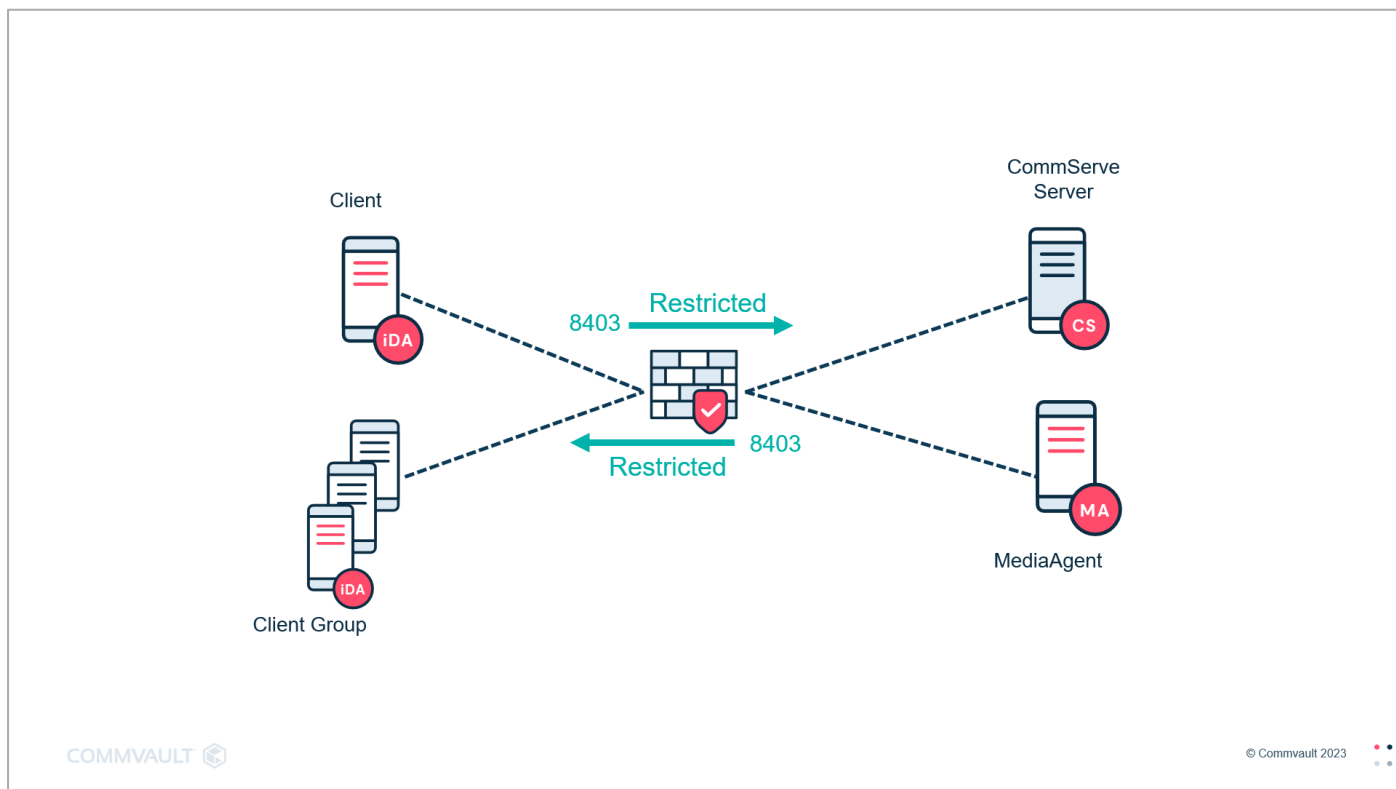
The only requirement is that the communication port (8400) and tunnel port (8403) are opened and accessible between the components.

Network topologies provide a template to deploy network route configurations to CommCell components. Once the simplified topology is configured, advanced network route settings remain available at the client group and client levels to further configure settings.

To use network topologies, client computers groups must be created first.

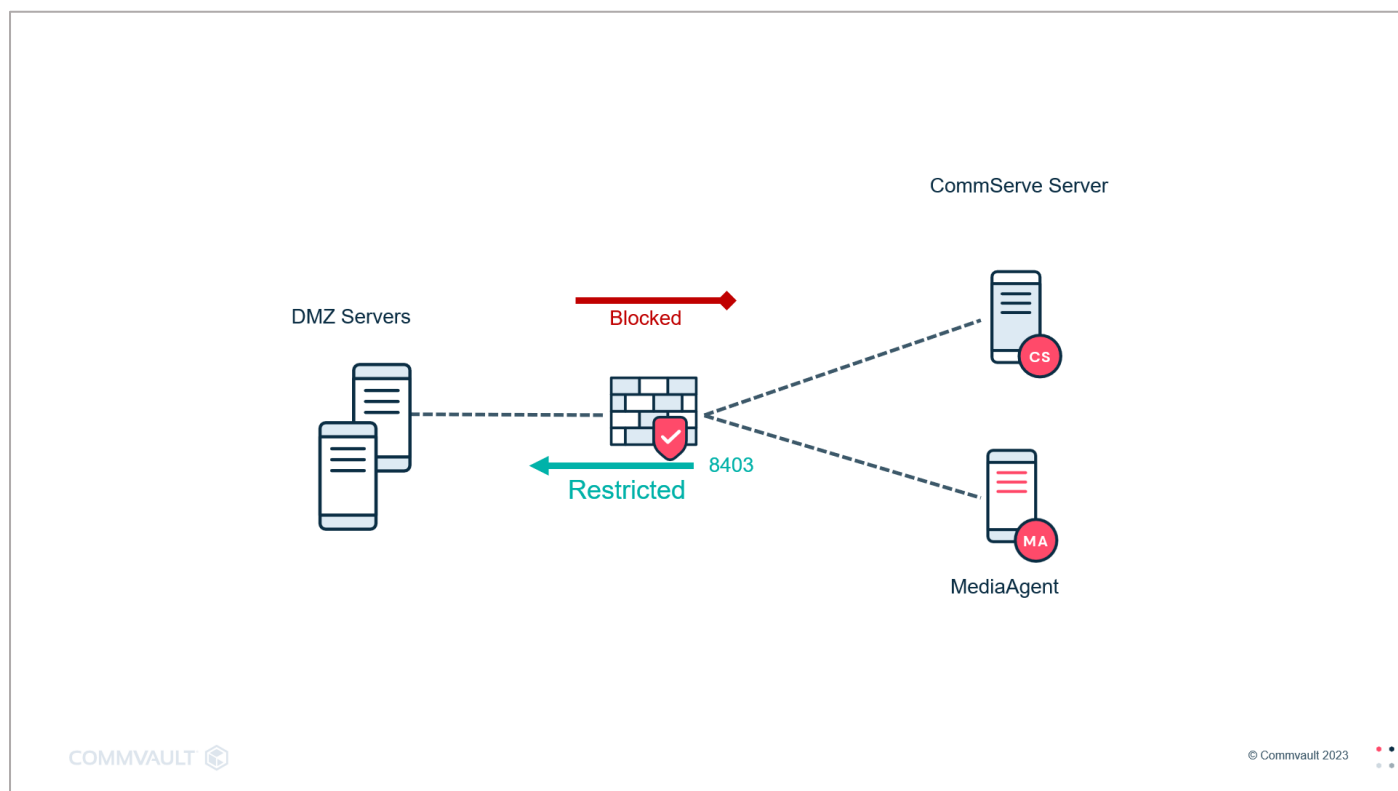
- **One-Way Network Topology** – Servers and DMZ servers
- **Two-Way Network Topology** – Servers and infrastructure machines
- **Network Gateway Topology** – Servers, infrastructure machines, and DMZ gateways
- **Cascaded Gateway Topology** – Servers, infrastructure machines, and DMZ gateways

Restricted Network Topology



A restricted network topology allows Commvault components to communicate through a firewall, but only on specific ports. A listening port is used to establish a connection between resources. In a restricted configuration, any resource can initiate communication. Before setting up network routes, the listening port must be opened bi-directionally on the firewall between components.

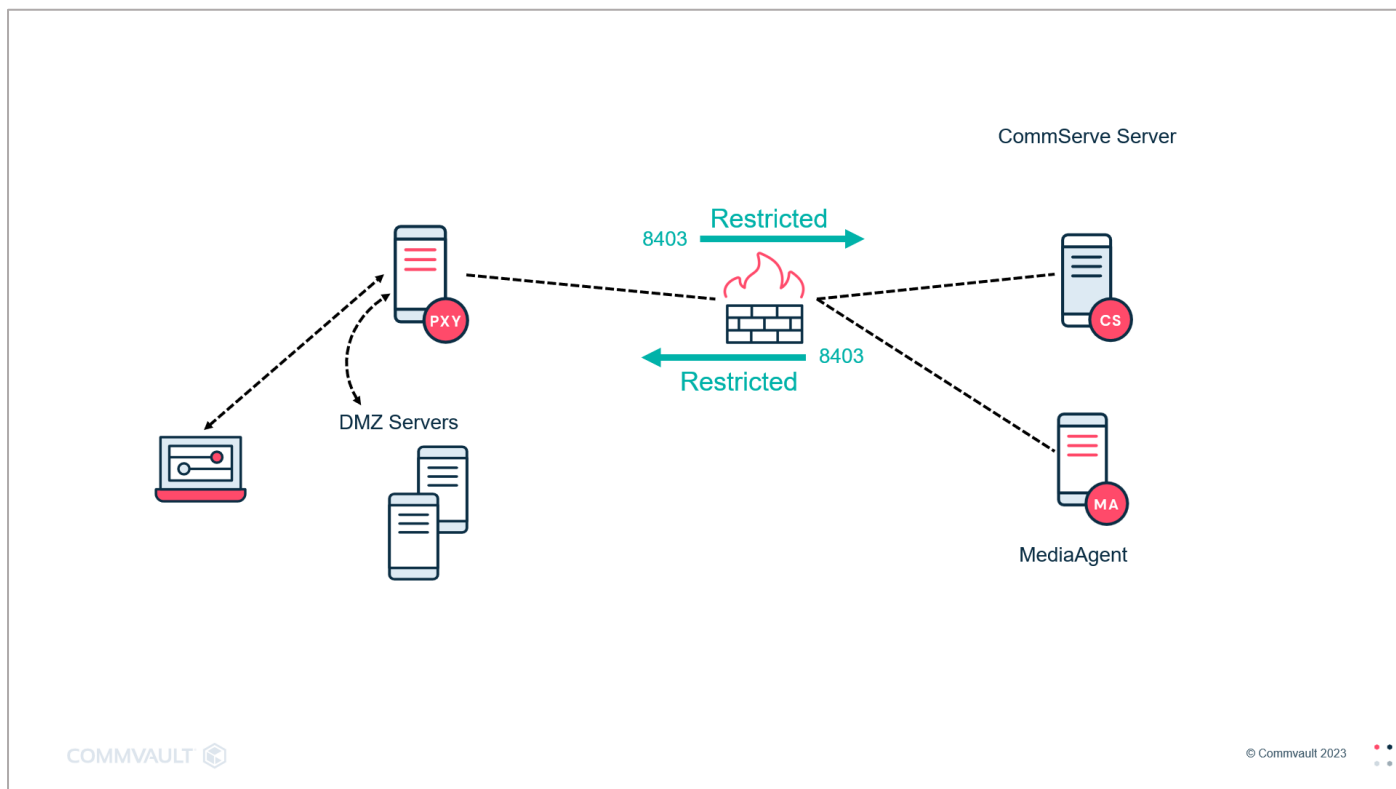
Blocked Network Topology



A blocked network route configuration restricts communication to be initiated by components on only one side of the firewall. This is typically used in a strongly secured DMZ or when external clients such as laptop clients are connecting to the CommCell environment from unsecured networks.

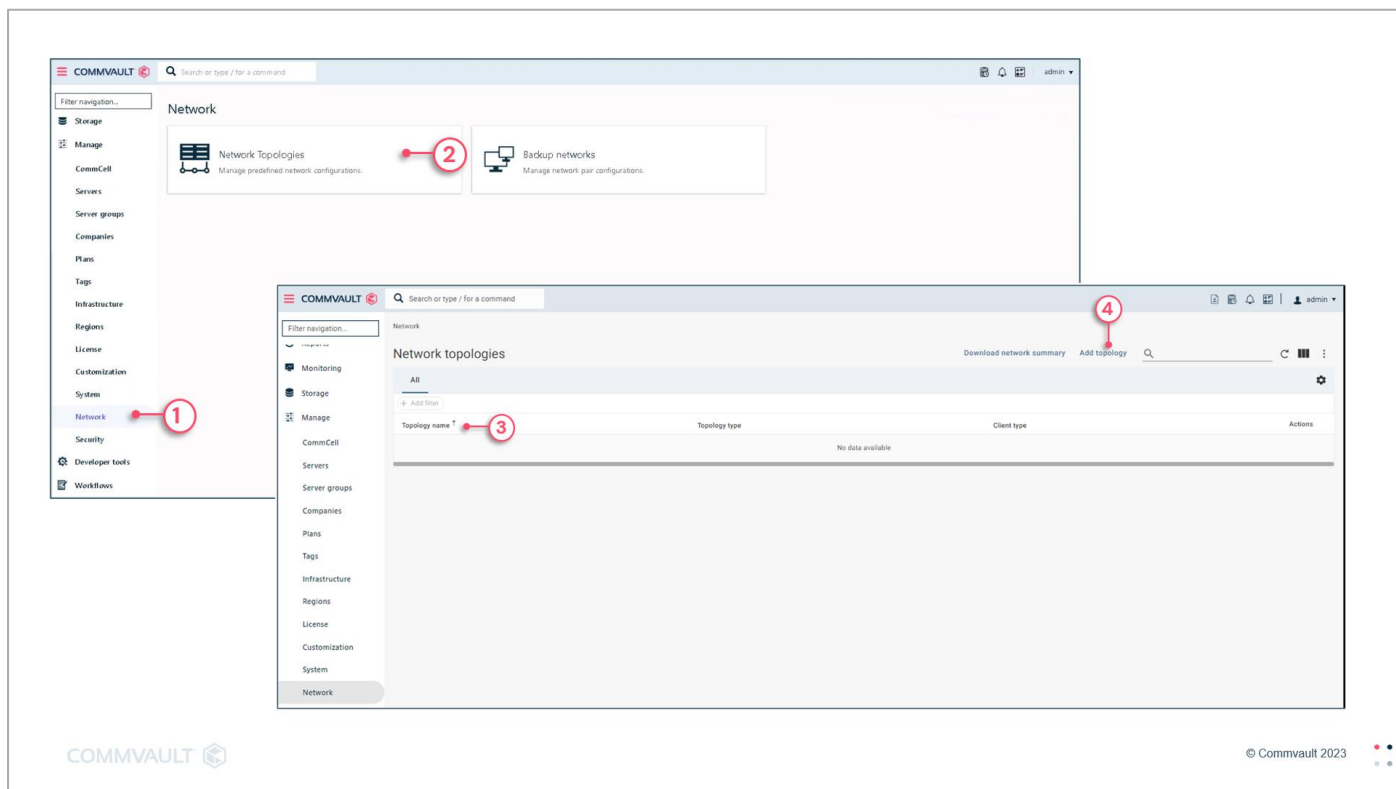
The CommCell component that establishes connection attempts to communicate with other CommCell resources when Commvault services start. This is managed in the *Options tab* of the firewall settings.

Network Gateway Topology



A Network Gateway route configuration is used when resources cannot directly communicate using a blocked or restricted connection. A network gateway is designated in the DMZ by selecting the Network Gateway Topology. Network routes must be configured from resources outside the firewall to the proxy and then from the proxy to resources inside the firewall.

Configure a Network Topology



Before configuring a topology, create the required server groups if needed. You can then launch the network topology configuration wizard.

1. Select **Manage > Network**
2. Click the **Network Topologies** tab
3. This window displays existing network topologies
4. Click **Add topology** to add a new topology

Configure a Network Topology ...cont'd

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Search or type / for a command

Filter navigation...

Network topologies

Monitoring

Storage

Manage

Developer tools

Workflows

Web console

Add Topology

General

Topology name* Intra-site Connection 5

Client type

☒ Servers 6

☐ Laptops

Topology type

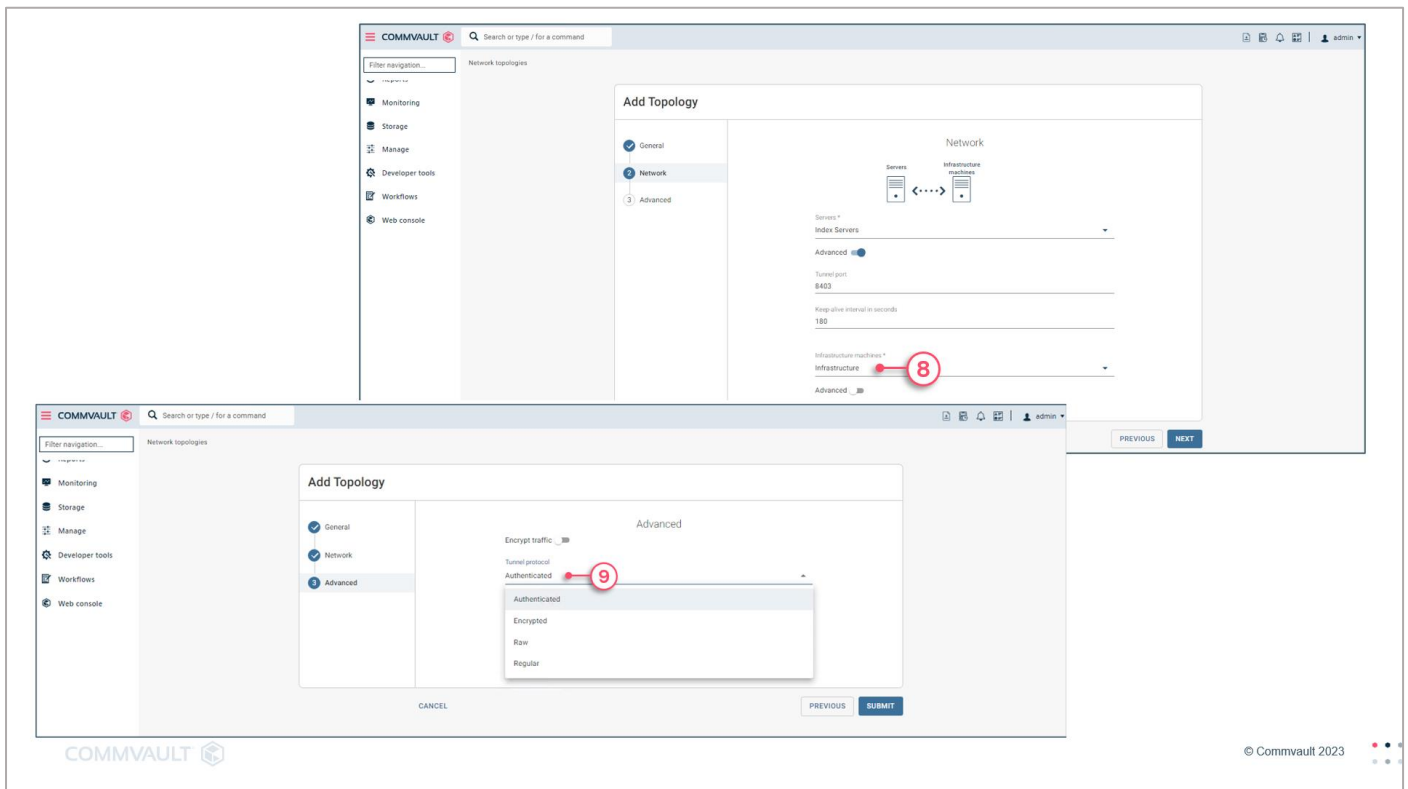
Two-way 7

Servers Infrastructure machines

CANCEL NEXT

5. Enter a name for the topology
6. Choose the client type
7. Choose the topology type based on your network configuration

Configure a Network Topology ...cont'd



8. Choose the required computer groups for which the topology will be applied
9. Configure any required advanced settings and click **Submit**